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**Lab 2**
**Practical 1.**

```
CREATE TABLE books (
    book_id INT PRIMARY KEY,
    title VARCHAR(50),
    author VARCHAR(50),
    price INT
);

INSERT INTO books VALUES
(1, 'alchemist', 'paulo', 350),
(2, '1984', 'george', 400),
(3, 'dune', 'frank', 600);

SELECT * FROM books;
```

```
+---------+-----------+--------+-------+
| book_id | title     | author | price |
+---------+-----------+--------+-------+
|       1 | alchemist | paulo  |   350 |
|       2 | 1984      | george |   400 |
|       3 | dune      | frank  |   600 |
+---------+-----------+--------+-------+
```

**Practical 2.**

```
CREATE TABLE players (
    player_id INT PRIMARY KEY,
    name VARCHAR(50),
    game VARCHAR(30),
    rank_level VARCHAR(20) DEFAULT 'bronze'
);

INSERT INTO players (player_id, name, game) VALUES
(11, 'noobmaster', 'valorant'),
(12, 'shadow', 'pubg');

SELECT * FROM players;
```

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| player_id | name | game | rank_level |
|---|---|---|---|
| 11 | noobmaster | valorant | bronze |
| 12 | shadow | pubg | bronze |

**Practical 3.**

```
CREATE TABLE flight (
   flight_no INT PRIMARY KEY,
   destination VARCHAR(50),
   gate_no VARCHAR(10)
);

INSERT INTO flight VALUES
(101, 'mumbai', 't1-a'),
(102, 'dubai', 't2-b'),
(103, 'london', 't3-c');

UPDATE flight
SET gate_no = 't2-g'
WHERE flight_no = 102;

SELECT * FROM flight;
```

| flight_no | destination | gate_no |
|---|---|---|
| 101 | mumbai | t1-a |
| 102 | dubai | t2-g |
| 103 | london | t3-c |

**Practical 4.**

```
CREATE TABLE movie (
```

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```
 movie_id INT PRIMARY KEY,
    name VARCHAR(50),
    genre VARCHAR(30),
    ticket_price INT
);

INSERT INTO movie VALUES
(21, 'interstellar', 'sci-fi', 250),
(22, 'oppenheimer', 'drama', 300),
(23, 'avatar', 'sci-fi', 350);

UPDATE movie
SET ticket_price = ticket_price + 50
WHERE genre = 'sci-fi';

SELECT * FROM movie;
```

```
+----------+--------------+--------+--------------+
| movie_id | name         | genre  | ticket_price |
+----------+--------------+--------+--------------+
|       21 | interstellar | sci-fi |          300 |
|       22 | oppenheimer  | drama  |          300 |
|       23 | avatar       | sci-fi |          400 |
+----------+--------------+--------+--------------+
```

**Practical 5.**

```
CREATE TABLE driver (
    driver_id INT PRIMARY KEY,
    name VARCHAR(50),
    vehicle_no VARCHAR(20),
    rating INT
);

INSERT INTO driver VALUES
(1, 'raj', 'ka-01-1234', 4),
(2, 'vicky', 'dl-02-5678', 3);

UPDATE driver
SET vehicle_no = 'ka-05-9999', rating = 5
WHERE driver_id = 1;
```

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SELECT * FROM driver;

| driver_id | name | vehicle_no | rating |
|---|---|---|---|
| 1 | raj | ka-05-9999 | 5 |
| 2 | vicky | dl-02-5678 | 3 |

**Practical 6.**

```
CREATE TABLE inventory (
   item_id INT PRIMARY KEY,
   item_name VARCHAR(50),
   quantity INT
);

INSERT INTO inventory VALUES
(51, 'monitor', 10),
(52, 'hdmi cable', 0),
(53, 'webcam', 15);

DELETE FROM inventory
WHERE item_id = 52;

SELECT * FROM inventory;
```

| item_id | item_name | quantity |
|---|---|---|
| 51 | monitor | 10 |
| 53 | webcam | 15 |

**Practical 7.**

```
CREATE TABLE train (
   train_no INT PRIMARY KEY,
   name VARCHAR(50),
   seats_left INT
```

```
);

INSERT INTO train VALUES
(1201, 'shatabdi', 0),
(1202, 'rajdhani', 45),
(1203, 'garib rath', 0);

DELETE FROM train
WHERE seats_left = 0;

SELECT * FROM train;
```

```
+----------+----------+------------+
| train_no | name     | seats_left |
+----------+----------+------------+
|     1202 | rajdhani |         45 |
+----------+----------+------------+
```

**Practical 8.**

```
CREATE TABLE temp_cart (
   cart_id INT PRIMARY KEY,
   item_name VARCHAR(50),
   price INT
);

INSERT INTO temp_cart VALUES
(1, 'chips', 20),
(2, 'coke', 40);

DELETE FROM temp_cart;

SELECT * FROM temp_cart;
```

> Output:
>
> Program did not output anything!

**Practical 9. ORDER BY**

```
CREATE TABLE Student (
   student_id INT PRIMARY KEY,
   name VARCHAR(50),
```

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```
  department VARCHAR(50),
   marks INT
);

INSERT INTO Student (student_id, name, department, marks) VALUES
(101, 'Mia', 'Computer Science', 85),
(102, 'Dia', 'Maths', 92),
(103, 'Hia', 'Computer Science', 78),
(104, 'Lia', 'Maths', 65),
(105, 'Sia', 'Computer Science', 95);

SELECT student_id, name, department, marks
FROM Student
ORDER BY marks DESC;
```

Output:

| student_id | name | department | marks |
|---|---|---|---|
| 105 | Sia | Computer Science | 95 |
| 102 | Dia | Maths | 92 |
| 101 | Mia | Computer Science | 85 |
| 103 | Hia | Computer Science | 78 |
| 104 | Lia | Maths | 65 |

**Practical 10**

```
CREATE TABLE Student (
   student_id INT PRIMARY KEY,
   name VARCHAR(50),
   department VARCHAR(50),
   marks INT
);

INSERT INTO Student (student_id, name, department, marks) VALUES
(101, 'Mia', 'Computer Science', 85),
(102, 'Dia', 'Maths', 92),
(103, 'Hia', 'Computer Science', 78),
(104, 'Lia', 'Maths', 65),
(105, 'Sia', 'Computer Science', 95);


SELECT department, COUNT(student_id) AS total_students, AVG(marks) AS average_marks
FROM Student
GROUP BY department;
```

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| department | total_students | average_marks |
|---|---|---|
| Computer Science | 3 | 86 |
| Maths | 2 | 78 |

**Practical 11 HAVING**

```
CREATE TABLE Student (
   student_id INT PRIMARY KEY,
   name VARCHAR(50),
   department VARCHAR(50),
   marks INT
);

INSERT INTO Student (student_id, name, department, marks) VALUES
(101, 'Mia', 'Computer Science', 85),
(102, 'Dia', 'Maths', 92),
(103, 'Hia', 'Computer Science', 78),
(104, 'Lia', 'Maths', 65),
(105, 'Sia', 'Computer Science', 95);

SELECT department, AVG(marks) AS average_marks
FROM Student
GROUP BY department
HAVING AVG(marks) > 80;
```

| department | average_marks |
|---|---|
| Computer Science | 86 |